

1. Tabulate the execution times of each of the individual approaches for computing distance in Python (i.e., run the shared code on your computer, note the times, and tabulate them).

Approach	For Loop	Apply Method	Vectorized
Time	0.0040 Seconds	0.0010 seconds	0.0000 seconds

2. Next, replicate the for-loop based approach (the first one) and two different ways to make that version more efficient, in R. Profile these three approaches, and tabulate the results.

Approach	For Loop	Apply Method	Vectorized
Time	0.02 Seconds	0.04 seconds	0.0000 seconds

3. Based on the computational efficiency of implementations in Python and R, which one would you prefer? Based on a consideration of implementation (i.e., designing and implementing the code), which approach would you prefer? Taking both of these (run time and coding time), which approach would you prefer?

Furthermore, Python presents better timing performance throughout all solution methods yet achieves 4x faster execution than R for the Apply method. The implementation of Python code becomes easier for general programmers because its syntax creates a natural flow which produces predictable results. The structured object-oriented framework of Python reduces development complexity because it integrates with detailed documentation and automatic error detection elements for code writing and debugging. The combination of performance excellence with straightforward code structure and straightforward implementation makes Python the better selection compared to other options.

4. Identify and describe one or two other considerations, in addition to these two, in determining which of the two environments – Python or R – is preferable to you

The vital elements to weigh for selecting Python or R comprise their maturity within connected frameworks and their ability to produce visual representations of data. The installation of Python benefits from extensive production system and web service connectivity and machine learning framework integration capabilities enabled by Flask, Django and TensorFlow libraries. The combination of deployment capabilities with production systems makes Python ideal for complete data science operations especially when designing deployable models. Both Python tools are proficient in visualization tasks but the ggplot2 package inside R receives particular praise for its easier graphical system and statistical capabilities which excel at data exploration

and scientific image creation. The decision could be affected by particular project needs and overall programming situations within which the code needs to operate.